



(51) International Patent Classification:

C07D 207/06 (2006.01)

(21) International Application Number:

PCT/US2024/017515

(22) International Filing Date:

27 February 2024 (27.02.2024)

(25) Filing Language:

English

(26) Publication Language:

English

(30) Priority Data:

63/487,446 28 February 2023 (28.02.2023) US

(71) Applicant: TARO PHARMACEUTICAL INDUSTRIES LTD. [IL/IL]; 14 Hakitor Street, P.O. Box 10347, 2624761 Haifa Bay (IL).

(71) Applicant (for EG only): TARO PHARMACEUTICALS U.S.A., INC. [US/US]; Three Skyline Drive, Hawthorne, New York 10532 (US).

(72) Inventors: OZER, Ilana; 7 Daliyot St., 2066524 Yokneam (IL). KAFTANOV, Yulia; 13 Kaspari St., Haifa (IL). SIMHON, Elliot; 7 Peretz Markish St., 3280507 Haifa (IL). DUSHKIN, Andrey; 119 Yefe Nof St., Haifa (IL). SHEFFER DEE-NOOR, Shani; 7 Got Levin St., 3292207 Haifa (IL). PIZEM, Hillel; 44 Deganya Road St., Netanya (IL). AVRAMOFF, Avi; 26 Lea St., Haifa (IL).

(74) Agent: WOODHOUSE, Scott M.; Medler Ferro Woodhouse & Mills PLLC, 8201 Greensboro Dr, Suite 1060, McLean, Virginia 22102 (US).

(81) Designated States (unless otherwise indicated, for every kind of national protection available): AE, AG, AL, AM, AO, AT, AU, AZ, BA, BB, BG, BH, BN, BR, BW, BY, BZ, CA, CH, CL, CN, CO, CR, CU, CV, CZ, DE, DJ, DK, DM, DO, DZ, EC, EE, EG, ES, FI, GB, GD, GE, GH, GM, GT, HN, HR, HU, ID, IL, IN, IQ, IR, IS, IT, JM, JO, JP, KE, KG, KH, KN, KP, KR, KW, KZ, LA, LC, LK, LR, LS, LU, LY, MA, MD, MG, MK, MN, MU, MW, MX, MY, MZ, NA, NG, NI, NO, NZ, OM, PA, PE, PG, PH, PL, PT, QA, RO, RS, RU, RW, SA, SC, SD, SE, SG, SK, SL, ST, SV, SY, TH, TJ, TM, TN, TR, TT, TZ, UA, UG, US, UZ, VC, VN, WS, ZA, ZM, ZW.

(84) Designated States (unless otherwise indicated, for every kind of regional protection available): ARIPO (BW, CV, GH, GM, KE, LR, LS, MW, MZ, NA, RW, SC, SD, SL, ST, SZ, TZ, UG, ZM, ZW), Eurasian (AM, AZ, BY, KG, KZ, RU, TJ, TM), European (AL, AT, BE, BG, CH, CY, CZ, DE, DK, EE, ES, FI, FR, GB, GR, HR, HU, IE, IS, IT, LT,

LU, LV, MC, ME, MK, MT, NL, NO, PL, PT, RO, RS, SE, SI, SK, SM, TR), OAPI (BF, BJ, CF, CG, CI, CM, GA, GN, GQ, GW, KM, ML, MR, NE, SN, TD, TG).

Published:

— with international search report (Art. 21(3))

(54) Title: METHODS OF PREPARING TRIFAROTENE

(57) Abstract: The present disclosure provides a process for the preparation of Trifarotene. The disclosure also provides for intermediates in the process described herein. Also provided are methods of making polymorph Form E of Trifarotene.

WO 2024/182418 A1

Methods of Preparing Trifarotene

FIELD OF THE INVENTION

[0001] The present disclosure relates to processes and intermediates for the synthesis of Trifarotene. Also provided is a novel polymorph of Trifarotene.

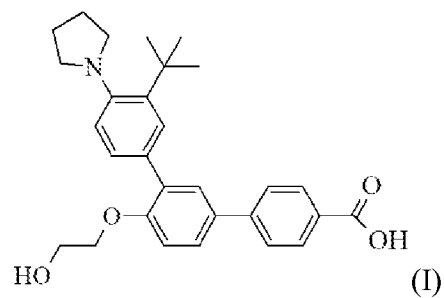
BACKGROUND

[0002] Retinoic acid receptor (RAR) selective compounds can treat acne, lamellar ichthyosis, photoaging and other diseases by modulating skin functions such as epidermal keratinization, differentiation, maturation, and proliferation. 3''-(tert-butyl)-4'-(2-hydroxyethoxy)-4''-(pyrrolidin-1-yl)-[1,1':3',1''-terphenyl]-4-carboxylic acid, commonly known as Trifarotene, is a potent and selective agonist of RAR- γ , the most common RAR found in the skin. Trifarotene is the latest fourth generation retinoid and was approved for the treatment of acne via topical administration by the Food and Drug Administration (FDA) in October 2019.

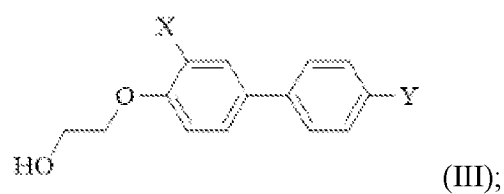
[0003] Current approaches for the synthesis of Trifarotene include several challenging steps and/or produce moderate to low yield Trifarotene. For example, described in WO 2006/066978, utilizes a reaction that is performed at -78°C and uses two separate protecting groups that must be hydrolyzed under different conditions. WO 2021/119351 uses a reaction in which all functional groups of the intermediates are hydrolyzed in one step. Multiple and complex reaction steps can decrease workflow efficiency and overall yield.

SUMMARY OF THE INVENTION

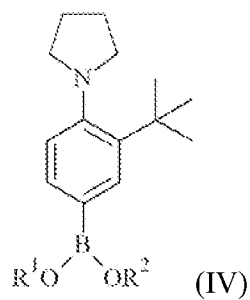
[0004] In some embodiments, the disclosure provides a process for the preparation of a compound of formula (I) [Trifarotene], or a salt thereof



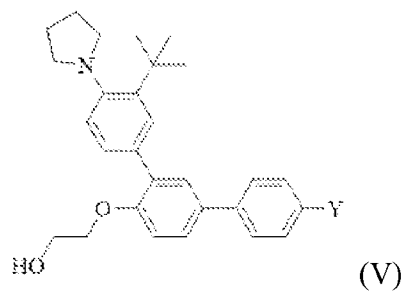
comprising reacting a compound of formula (III)



with a compound of formula (IV)



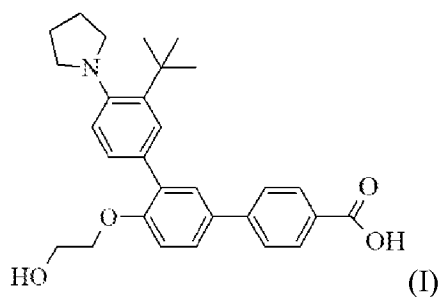
to obtain a compound of formula (V)



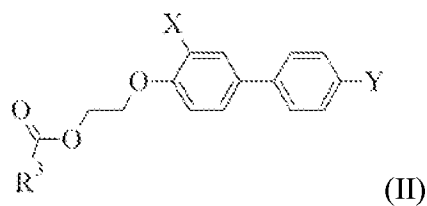
and hydrolyzing the compound of formula (V) to obtain the compound of formula (I), wherein R^1 and R^2 are independently hydrogen or a linear or branched C_1 - C_3 alkyl, wherein R^1 and R^2 can be

the same or different or R^1 and R^2 together form a pinacolate in the presence of a catalyst, wherein X is a halogen or triflate, and wherein Y is a nitrile (CN) or amide (CONH₂).

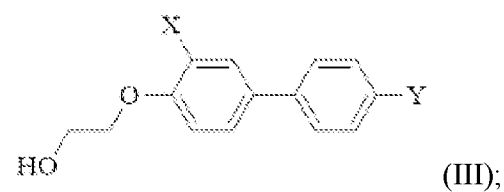
[0005] In some embodiments, the disclosure provides a process for the preparation of a compound of formula (I) [Trifarotene], or a salt thereof



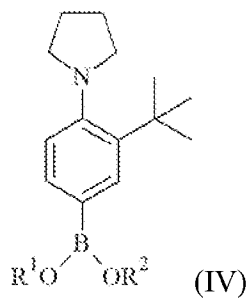
comprising hydrolyzing a compound of formula (II)



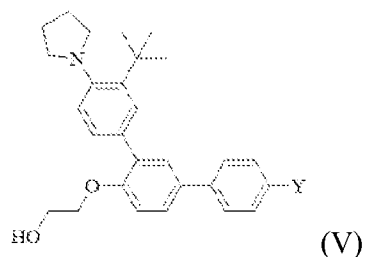
in the presence of a base, to obtain a compound of formula (III)



reacting the compound of formula (III) with a compound of formula (IV)



to obtain a compound of formula (V)



and hydrolyzing the compound of formula (V) to obtain the compound of formula (I), wherein R^1 and R^2 are independently hydrogen or a linear or branched C_1 - C_3 alkyl, wherein R^1 and R^2 can be the same or different; or R^1 and R^2 together form a pinacolate in the presence of a catalyst, wherein R^3 is hydrogen, a substituted or unsubstituted linear or branched C_1 - C_8 alkyl, a substituted or unsubstituted linear or branched C_1 - C_8 alkenyl group, a substituted or unsubstituted linear or branched C_1 - C_8 alkynyl group, a substituted or unsubstituted cycloalkyl group, a substituted or unsubstituted aryl group, a substituted or unsubstituted heterocycle, a substituted or unsubstituted heteroaryl, or a substituted or unsubstituted C_1 - C_8 alkyl group comprising a heteroatom, wherein X is a halogen or triflate, and wherein Y is a nitrile (CN) or amide (CONH₂).

[0006] In some embodiments, the X is iodine.

[0007] In some embodiments, the hydrolysis is performed in the presence of a solvent comprising water, methanol (MeOH), ethanol (EtOH), propanol (PrOH), isopropanol (IPA), or any mixture thereof. In some embodiments, the solvent comprises water and ethanol.

[0008] In some embodiments, the reaction is performed in the presence of a solvent comprising toluene, Dimethylacetamide (DMA) dimethylformamide (DMF), dimethyl sulfoxide (DMSO), tetrahydrofuran (THF), dioxane, n-butanol (n-BuOH), isopropanol (IPA), dimethyl ether (DME), diethyl ether, or any mixture thereof.

[0009] In some embodiments, the reaction is performed in the presence of a base comprising K_2CO_3 , CH_3CO_2K , K_3PO_4 , $KOtBu$, Na_2CO_3 , $NaHCO_3$, $NaOMe$, $CaCO_3$, Li_2CO_3 , Cs_2CO_3 , Ag_3PO_4 , Ag_2O , Tl_2CO_3 , $TIOEt$, $TIOH$, $t-BuNH_2$, KOH , $NaOH$, $LiOH$, $Ba(OH)_2$, or combination thereof.

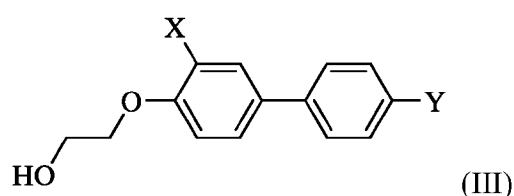
[0010] In some embodiments, the catalyst comprises a metal selected from Pd, Cu, or Ni. In some embodiments, the catalyst comprises at least two atoms of the metal. In some embodiments, the catalyst is a Pd catalyst selected from Pd(PPh₃)₂Cl₂ [bis(triphenylphosphine)palladium(II) dichloride]; Pd(PPh₃)₄ [tetrakis(triphenylphosphine)palladium(0)]; Pd(OAc)₂ [palladium(II) diacetate]; XPhos Pd-G3 [(2-dicyclohexylphosphino-2',4',6'-triisopropyl-1,1'-biphenyl)[2-(2'-amino-1,1'-biphenyl)]palladium(II) methanesulfonate]; SPhos-Pd-G2 [chloro(2-dicyclohexylphosphino-2',6'-dimethoxy-1,1'-biphenyl)[2-(2'-amino-1,1'-biphenyl)]palladium(II)]; CATACXIUM® A Pd G3 (mesylate[(di(1-adamantyl)-n-butylphosphine)-2-(2'-amino-1,1'-biphenyl)]palladium(II) or [(di(1-adamantyl)-butylphosphine)-2-(2'-amino-1,1'-biphenyl)]palladium(II) methanesulfonate); APhos Pd G3 (palladium G3-(4-(N,N-dimethylamino)phenyl)di-tert-butylphosphine] or [4-(di-tert-butylphosphino)-N,N-dimethylaniline-2-(2'-aminobiphenyl)]palladium(II) methanesulfonate); P(Cy₃) Pd-G3 (palladium G3-tricyclohexylphosphine or [(tricyclohexylphosphine)-2-(2'-aminobiphenyl)]palladium(II) methanesulfonate); Allylpalladium(II) chloride dimer (bis(allyl)dichlorodipalladium); or Pd(dppf)Cl₂ [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium(II)].

[0011] In some embodiments, the compounds of formula (III) and formula (IV) are present in a molar ratio of about 1:10 to about 10:1, preferably about 1:5 to about 5:1, more preferably about 1:1. In some embodiments, the compounds of formula (III) and formula (IV) are independently present in an amount of about 0.01 to about 1 mol/L (solvent), preferably about 0.05 to about 0.5 mol/L (solvent), more preferably about 0.1 to about 0.4 mol/L (solvent).

[0012] In some embodiments, the catalyst is present at about 0.001 to about 1 molar equivalents relative to the compounds of formula (III) or formula (IV), preferably about 0.002 to about 0.5 molar equivalents relative to the compounds of formula (III) or formula (IV), more preferably about 0.003 to about 0.1 molar equivalents relative to the compounds of formula (III) or formula (IV). In some embodiments, the base is present at about 0.1 to about 10 molar equivalents relative to the compounds of formula (III) or formula (IV), preferably about 1 to about 6 molar equivalents relative to the compounds of formula (III) or formula (IV), more preferably about 2 to about 4 molar equivalents relative to the compounds of formula (III) or formula (IV).

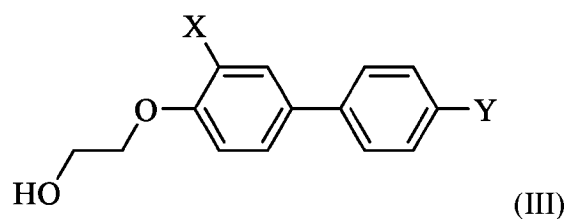
[0013] In some embodiments, the pH is adjusted to about 3.4 to about 6.0 after hydrolyzing the compound of formula (V). In some embodiments, the pH is adjusted to about 4.0 to about 6.0 after hydrolyzing the compound of formula (V). In some embodiments, the pH is adjusted to about 4.5 to about 6.0 after hydrolyzing the compound of formula (V). In some embodiments, the pH is adjusted to about 5.0 to about 6.0 after hydrolyzing the compound of formula (V). In some embodiments, the pH is adjusted to 5.5 after hydrolyzing the compound of formula (V).

[0014] In some embodiments, the disclosure provides a compound of formula (III)

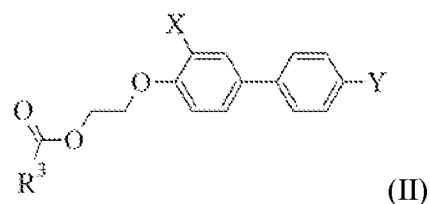


wherein X is a halogen or triflate, and wherein Y is a nitrile (CN) or amide (CONH₂).

[0015] In some embodiments, the disclosure provides a process for the preparation of a compound of formula (III)



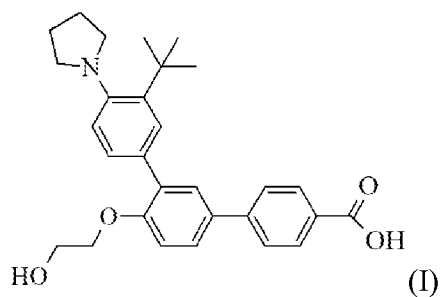
comprising hydrolyzing a compound of formula (II)



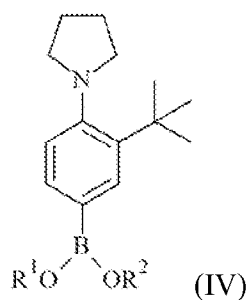
wherein R³ is hydrogen, a substituted or unsubstituted linear or branched C₁-C₈ alkyl, a substituted or unsubstituted linear or branched C₁-C₈ alkenyl group, a substituted or unsubstituted linear or branched C₁-C₈ alkynyl group, a substituted or unsubstituted cycloalkyl group, a substituted or

unsubstituted aryl group, a substituted or unsubstituted heterocycle, a substituted or unsubstituted heteroaryl, or a substituted or unsubstituted C₁-C₈ alkyl group comprising a heteroatom, wherein X is a halogen or triflate, and wherein Y is a nitrile (CN) or amide (CONH₂) in the presence of a base.

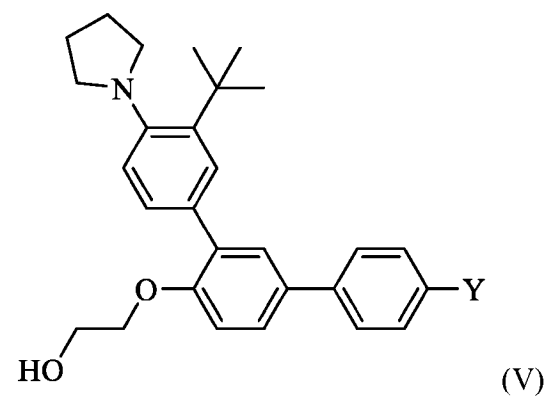
[0016] In some embodiments, the process further comprises further comprising preparing a compound of formula (I) [Trifarotene], or a salt thereof



by reacting the compound of formula (III) with a compound of formula (IV)



to obtain a compound of formula (V)



wherein R^1 and R^2 are independently hydrogen or a linear or branched C_1 - C_3 alkyl, wherein R^1 and R^2 can be the same or different; or R^1 and R^2 together form a pinacolate in the presence of a catalyst, wherein Y is a nitrile (CN) or amide (CONH₂) and hydrolyzing the compound of formula (V) to obtain the compound of formula (I).

[0017] In some embodiments, the disclosure provides a Form E polymorph of Trifarotene, comprising providing Trifarotene according to a process described herein and suspending the Trifarotene in ethyl acetate to obtain a Form E polymorph of Trifarotene.

[0018] In some embodiments, the disclosure provides a Form E polymorph of the compound of Formula (I) [Trifarotene], wherein the Form E polymorph shows an X-ray powder diffraction pattern having characteristic peaks at reflection angle 2θ of 3.8 ± 0.2 , 7.4 ± 0.2 , 8.9 ± 0.2 , 10.9 ± 0.2 , 13.1 ± 0.2 , 14.6 ± 0.2 , 16.6 ± 0.2 , 18.2 ± 0.2 , 22.3 ± 0.2 , and 24.4 ± 0.2 degrees. In some embodiments, any single impurity in the Form E polymorph is less than 0.15%.

BRIEF DESCRIPTION OF THE DRAWINGS

[0019] FIG. 1 shows an exemplary process for the preparation of Trifarotene [Formula (I)] as described in embodiments herein.

[0020] FIG. 2 is the XRD spectrum of Trifarotene polymorph Form E.

DETAILED DESCRIPTION OF THE INVENTION

[0021] The present disclosure relates to methods for the preparation of Trifarotene. The methods provided herein advantageously simplify the preparation process by reducing or eliminating reaction steps. In some embodiments, the methods described herein provide for improved methods of making polymorph Form E of Trifarotene. In some embodiments, the methods described herein provide for Trifarotene of increased purity.

[0022] As used herein, “a” or “an” may mean one or more. As used herein, when used in conjunction with the word “comprising,” the words “a” or “an” may mean one or more than one. As used herein, “another” or “a further” may mean at least a second or more.

[0023] Throughout this application, the term “about” is used to indicate that a value includes the inherent variation of error for the method/device being employed to determine the value, or the variation that exists among the study subjects. Typically, the term “about” is meant to encompass approximately or less than 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 11%, 12%, 13%, 14%, 15%, 16%, 17%, 18%, 19% or 20% or higher variability, depending on the situation. In some embodiments, one of skill in the art will understand the level of variability indicated by the term “about,” due to the context in which it is used herein. It should also be understood that use of the term “about” also includes the specifically recited value.

[0024] The use of the term “or” in the claims is used to mean “and/or,” unless explicitly indicated to refer only to alternatives or the alternatives are mutually exclusive, although the disclosure supports a definition that refers to only alternatives and “and/or.”

[0025] As used herein, the terms “comprising” (and any variant or form of comprising, such as “comprise” and “comprises”), “having” (and any variant or form of having, such as “have” and “has”), “including” (and any variant or form of including, such as “includes” and “include”) or “containing” (and any variant or form of containing, such as “contains” and “contain”) are inclusive or open-ended and do not exclude additional, unrecited elements or method steps. It is contemplated that any embodiment discussed in this specification can be implemented with respect to any compound or method of making as described in the present disclosure. Furthermore, the compounds of the present disclosure can be used to achieve and optimize the methods of making of the present disclosure.

[0026] The use of the term “for example” and its corresponding abbreviation “e.g.” (whether italicized or not) means that the specific terms recited are representative examples and embodiments of the disclosure that are not intended to be limited to the specific examples referenced or cited unless explicitly stated otherwise.

[0027] As used herein, “between” is a range inclusive of the ends of the range. For example, a number between x and y explicitly includes the numbers x and y, and any numbers that fall within x and y.

terphenyl]-4-carboxylic acid, which was suspended in ethyl acetate (1000 ml) followed by filtration of the product. The white wet solid of 3''-(tert-butyl)-4'-(2-hydroxyethoxy)-4''-(pyrrolidin-1-yl)-[1,1':3',1''-terphenyl]-4-carboxylic acid was suspended in water (4000 mL) and heated to 70 to 100°C for 5 to 9 hours and filtered off. The white wet powder was then dried under vacuum to afford 120 g of 3''-(tert-butyl)-4'-(2-hydroxyethoxy)-4''-(pyrrolidin-1-yl)-[1,1':3',1''-terphenyl]-4-carboxylic acid pure [Trifarotene; Formula I]. 2-((3''-(tert-butyl)-4-cyano-4''-(pyrrolidin-1-yl)-[1,1':3',1''-terphenyl]-4'-yl)oxy) ethanol was analyzed by HPLC analytical method.

Example 2: Comparison of the methods of the present disclosure to previously described methods

[0073] The total yield from the methods of the present disclosure were compared to the yield of a previously known method of making Trifarotene, as found in US 17/756,994. The previously known method combined a Trifarotene nitrile ether precursor (e.g., 4-((4-bromo-2-(tert-butyl)phenyl)amino)-4-oxobutanoic acid, “Trif-nitrile ether”), with a Trifarotene borate precursor (e.g., (3-(tert-butyl)-4-pyrrolidine-1-yl)phenyl)boronic acid, “Trif-Boronic”) to form a Trifarotene nitrile ester (referred to herein as “the Trif-nitrile ester pathway”). The present disclosure provides a new method of synthesis in which the Trifarotene nitrile ether precursor is first converted to a Trifarotene alcohol precursor (Formula III).

[0074] HPLC analysis was conducted on the samples using both the old pathway (the Trif-nitrile ester pathway) and the new pathway (Trif-alcohol ether pathway). A RP18 column, was used. Using gradient of water: 0.02M Ammonium acetate:acetonitrile, with a flow rate of 1.0 mL/min.

[0075] A comparison of the yields resulting from the two pathways is presented in **Table 1**.

Table 1

Name	Trif-nitrile ester pathway		Trif-nitrile alcohol (%)		
	Trial #1	Trial #2	Trial #1	Trial #2	Trial #3
Yield, %	30.8	18.5	63.8	68	66.0

[0076] The data demonstrates that the Trif-alcohol ether pathway described herein results in a significant increase in yield relative to the previously used method, while maintaining good purity.

Example 3. Preparation of Trifarotene - Form E Polymorph and Its Analysis

[0077] 150 mg Trifarotene obtained in Example 1 was suspended in 5 mL of MeOH or EtOAc and shaken at 300 rpm for 2 days at room temperature. The product was filtered off and dried under ambient conditions for 2-6 days (3 days). The resulting crystal form is the Form E polymorph as determined by XRPD and IR spectrum analysis.

[0078] An exemplary diffraction pattern of the Trifarotene Form E polymorph can be found in Figure 2. **Table 2** lists the main X-Ray powder diffraction peaks. The XRD is accordance with what is described in US 17/756,994, incorporated herein in its entirety.

Table 2: X-Ray powder main diffraction peaks ($^{\circ}2\theta$) of Trifarotene

$^{\circ}2\theta$
3.8
7.4
8.9
10.9
13.1
14.6
16.6
18.2
22.3
24.4

[0079] The samples were analyzed by HPLC analytical method described above. **Table 3** shows the HPLC purity results of the Form E polymorph of Trifarotene.

Table 3

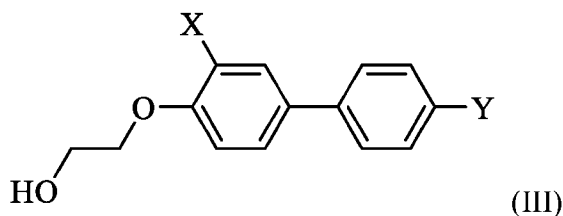
Sample name	Ratio Trifarotene: organic solvent: water	HPLC purity results (%) RT (min)									
		12.6 min	18.5 min	20.7 min	29.3 min	30.2 min	31.8 min	32.8 min	33.9 min	34.1 min	43.7 min
1	MeOH 1:10:1	0.02	0.04	0.06	0.01	99.58	0.02	0.12			0.13
2	EtOAc 1:10:1	0.02	0.05	0.03		99.75	0.03	0.08			
3	EtOAc 1:10:1	0.02	0.04	0.03		99.80	0.02	0.07			
4	EtOAc 1:10:1	0.01	0.06	0.04		99.76	0.03	0.08	0.02		
5	EtOH 1:10:1	0.02	0.04	0.08	0.02	99.64	0.03	0.12	0.02	0.01	
6	EtOH 1:10:1	0.03	0.03	0.07	0.02	99.64	0.03	0.12			0.06
7	EtOH 1:10:1 +seeding with polymorph E (10%)	0.02	0.04	0.05	0.02	99.71	0.03	0.11	0.02		

[0080] The purity level of Trifarotene Form E polymorphs (as determined by HPLC prepared in EtOAc 1:10:1 (samples 2, 3 and 4) was higher than the purity level observed for the Form E polymorphs prepared in MeOH 1:10:1 (sample 1) and EtOH 1:10:1 (sample 5, 6 and 7). See the

Trifarotene peak at 30.2 min (bolded). These results indicate that the preparation of Trifarotene - Form E Polymorph using ethyl acetate provides Trifarotene with higher purity compared to the other solvents tested. Furthermore, all impurities in Trifarotene Form E Polymorph suspended in ethyl acetate are less than 0.15%.

7. The process according to any one of claims 1 to 6, wherein the reaction is performed in the presence of a base comprising K_2CO_3 , CH_3CO_2K , K_3PO_4 , $KOtBu$, Na_2CO_3 , $NaHCO_3$, $NaOMe$, $CaCO_3$, Li_2CO_3 , Cs_2CO_3 , Ag_3PO_4 , Ag_2O , Tl_2CO_3 , $TlOEt$, $TlOH$, $t-BuNH_2$, KOH , $NaOH$, $LiOH$, $Ba(OH)_2$, or combination thereof.
8. The process according to any one of claims 1 to 7, wherein the catalyst comprises a metal selected from Pd, Cu, or Ni.
9. The process according to claim 8, wherein the catalyst comprises at least two atoms of the metal.
10. The process according to claim 8, wherein the catalyst is a Pd catalyst selected from:
 - (i) $Pd(PPh_3)_2Cl_2$ [Bis(triphenylphosphine)palladium(II) dichloride];
 - (ii) $Pd(PPh_3)_4$ [Tetrakis(triphenylphosphine)palladium(0)];
 - (iii) $Pd(OAc)_2$ [Palladium (II) diacetate];
 - (iv) XPhos Pd-G3 [(2-Dicyclohexylphosphino-2',4',6'-triisopropyl-1,1'-biphenyl)[2-(2'-amino-1,1'-biphenyl)]palladium(II) Methanesulfonate];
 - (v) SPhos-Pd-G2 [Chloro(2-dicyclohexylphosphino-2',6'-dimethoxy-1,1'-biphenyl)[2-(2'-amino-1,1'-biphenyl)]palladium(II)];
 - (vi) CATACXIUM® A Pd-G3 [Mesylate[(di(1-adamantyl)-n-butylphosphine)-2-(2'-amino-1,1'-biphenyl)]palladium(II), [(Di(1-adamantyl)-butylphosphine)-2-(2'-amino-1,1'-biphenyl)]palladium(II) Methanesulfonate];
 - (vii) APhos-Pd-G3 [Palladium G3-(4-(N,N-Dimethylamino)phenyl)di-tert-butylphosphine, [4-(Di-tert-butylphosphino)-N,N-dimethylaniline-2-(2'-aminobiphenyl)]palladium(II) Methanesulfonate];
 - (viii) $P(Cy_3) Pd-G3$ [(Tricyclohexylphosphine)-2-(2'-aminobiphenyl)]palladium(II) Methanesulfonate];
 - (ix) Allylpalladium(II) chloride dimer Bis(allyl)dichlorodipalladium; or
 - (x) $Pd(dppf)Cl_2$ [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium(II)].
11. The process according to any one of claims 1 to 10, wherein the compounds of formula (III) and formula (IV) are present in a molar ratio of about 1:10 to about 10:1, preferably about 1:5 to about 5:1, more preferably about 1:1.

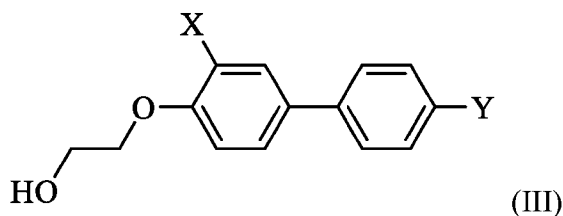
12. The process according to any one of claims 1 to 10, wherein the compounds of formula (III) and formula (IV) are independently present in an amount of about 0.01 to about 1 mol/L (solvent), preferably about 0.05 to about 0.5 mol/L (solvent), more preferably about 0.1 to about 0.4 mol/L (solvent).
13. The process according to any one of claims 1 to 12, wherein the catalyst is present at about 0.001 to about 1 molar equivalents relative to the compounds of formula (III) or formula (IV), preferably about 0.002 to about 0.5 molar equivalents relative to the compounds of formula (III) or formula (IV), more preferably about 0.003 to about 0.1 molar equivalents relative to the compounds of formula (III) or formula (IV).
14. The process according to claim 7, wherein the base is present at about 0.1 to about 10 molar equivalents relative to the compounds of formula (III) or formula (IV), preferably about 1 to about 6 molar equivalents relative to the compounds of formula (III) or formula (IV), more preferably about 2 to about 4 molar equivalents relative to the compounds of formula (III) or formula (IV).
15. The process according to any one of claims 1 to 13, wherein the pH is reduced to about 3.4 to about 6.0 after hydrolyzing the compound of formula (V).
16. A compound of formula (III)



wherein X is a halogen or triflate; and

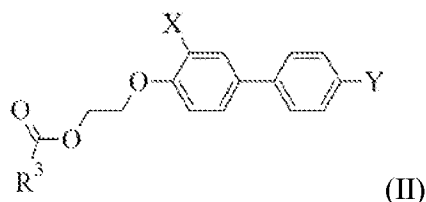
wherein Y is a nitrile (CN) or amide (CONH₂).

17. A process for the preparation of a compound of formula (III)



comprising

hydrolyzing a compound of formula (II)



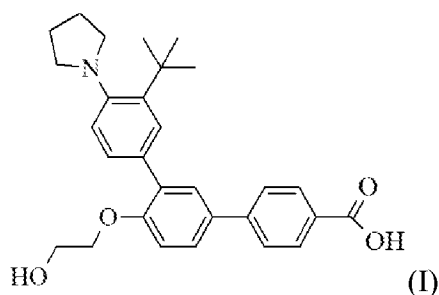
wherein R³ is hydrogen, a substituted or unsubstituted linear or branched C₁-C₈ alkyl, a substituted or unsubstituted linear or branched C₁-C₈ alkenyl group, a substituted or unsubstituted linear or branched C₁-C₈ alkynyl group, a substituted or unsubstituted cycloalkyl group, a substituted or unsubstituted aryl group, a substituted or unsubstituted heterocycle, a substituted or unsubstituted heteroaryl, or a substituted or unsubstituted C₁-C₈ alkyl group comprising a heteroatom;

wherein X is a halogen or triflate; and

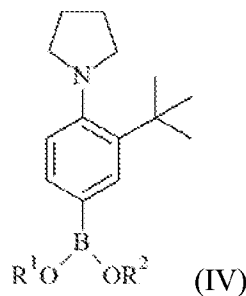
wherein Y is a nitrile (CN) or amide (CONH₂),

in the presence of a base.

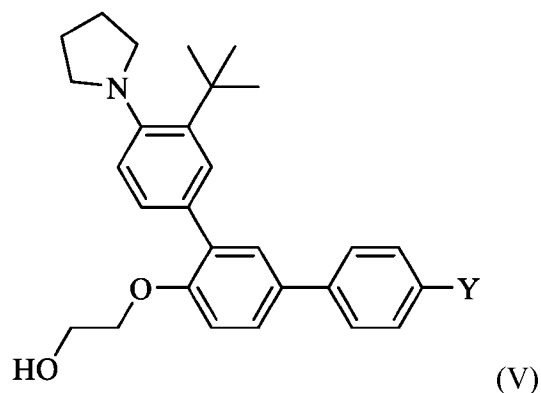
18. The process of claim 17, further comprising preparing a compound of formula (I) [Trifarotene], or a salt thereof



by reacting the compound of formula (III) with a compound of formula (IV)



to obtain a compound of formula (V)



wherein R^1 and R^2 are independently hydrogen or a linear or branched C_1 - C_3 alkyl, wherein R^1 and R^2 can be the same or different; or R^1 and R^2 together form a pinacolate in the presence of a catalyst,

wherein Y is a nitrile (CN) or amide (CONH₂); and

- c) hydrolyzing the compound of formula (V) to obtain the compound of formula (I).
19. The method of claim 18, wherein the pH is adjusted to a pH of about 3.4 to about 6.0 after the hydrolysis of (c).
 20. A process for preparing a Form E polymorph of Trifarotene, comprising:
 - a) providing Trifarotene according to the process of any one of claims 1 to 19;
 - b) suspending the Trifarotene in ethyl acetate to obtain a Form E polymorph of Trifarotene.
 21. The process of claim 20, wherein any single impurity is less than 0.15%.

22. The process of claim 21, wherein an X-ray powder diffraction (XRPD) comprising three or more 2θ values selected from 3.8 ± 0.2 , 7.4 ± 0.2 , 8.9 ± 0.2 , 10.9 ± 0.2 , 13.1 ± 0.2 , 14.6 ± 0.2 , 16.6 ± 0.2 , 18.2 ± 0.2 , 22.3 ± 0.2 , and 24.4 ± 0.2 .

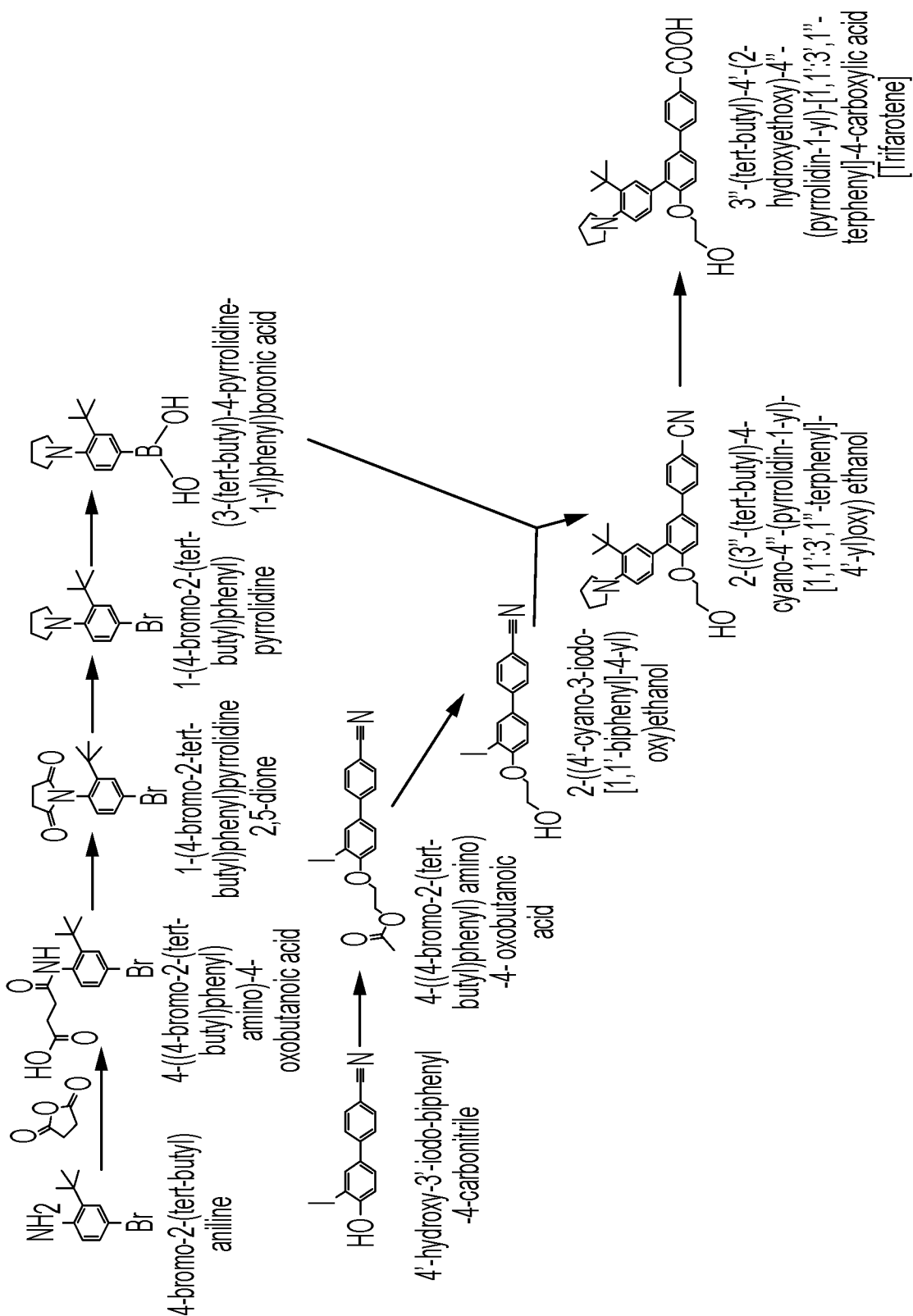


FIG. 1

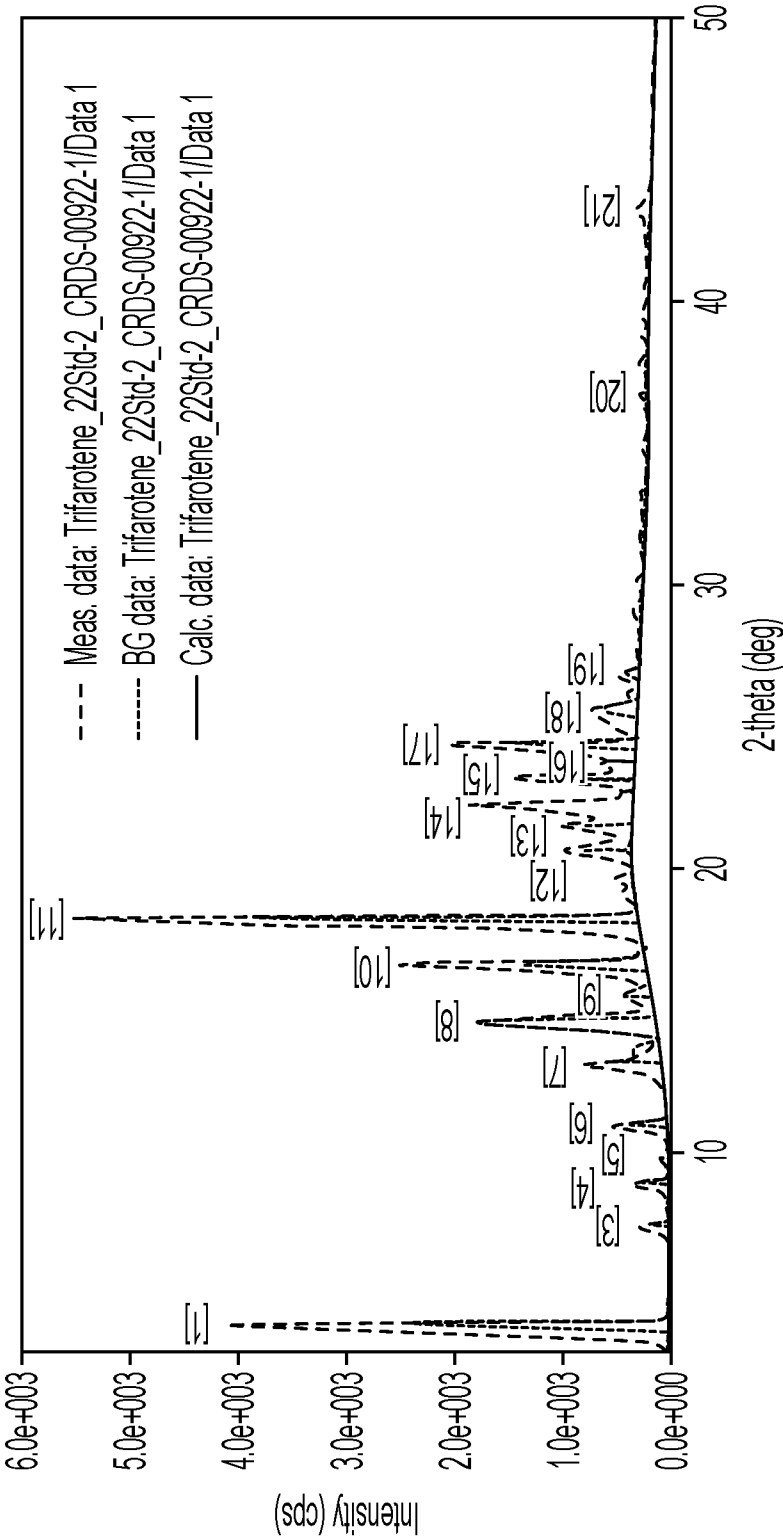


FIG. 2